

ASTRALINK — AI-BASED VIRTUAL CAMERA TRACKING SYSTEM FOR MOBILE FSOC

live simulation

Coarse Alignment - Performance Report

UTC 2026-09-27T08:48:46

1. RUN CONFIGURATION

Frames Processed

253

Duration

18.19 s

Processing FPS

153.9

Avg Processing

6.50 ms

Max Processing

7.81 ms

2. TRACKING PERFORMANCE

Acquisition Time

0.60 s

<= 2 s

Average Tracking Error

1.98 px

Maximum Tracking Error

4.00 px

<= 10 px

RMSE Tracking Error

2.06 px

Centroid Error Avg/Max

0.29 / 0.50 px

ground truth only

Lock Retention

100.00 %

Target Loss

0.00 %

< 5 %

Re-acquisition Avg/Max

—

<= 1 s

Successful Re-acq.

0

0 loss events

3. PS REQUIREMENT CHECK

Parameter	Measured	Requirement	Status
Acquisition Time	0.60	<= 2.0 s	PASS
Tracking Error (Max)	4.00	<= 10 px	PASS
Center Error (Final)	4.00	<= 10 px	PASS
Target Loss	0.00 %	< 5%	PASS
Re-acquisition Time	—	<= 1.0 s	PENDING
Processing Speed	153.9 FPS	>= 20 FPS	PASS

Ground truth was available: centroid accuracy is measured.

OVERALL: PENDING

Source: live simulation | Frames: 253 | FPS: 153.9